Repository Appendix C: Extended theoretical and methodological details

Supplement to Benchmark-Gated Composite Mean Estimation

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Repository Appendix C: Extended theoretical and methodological details

C.1 Regime-dependent estimator performance

Classical location estimators offer different balances between efficiency and robustness, and these balances vary across distributional regimes. Huber's ε contamination model formalizes how even a small proportion of gross errors can distort empirical summaries, particularly those based on moments [huber1964robust]. An estimator that is efficient under clean Gaussian sampling may perform poorly under right skewness, heavy-tailed noise, or adversarial contamination. Recent concentration results for heavy-tailed and structured settings reinforce this dependence on the underlying regime [oliveira2022improved,cai2022nonasymptotic]. The objective is to determine which estimator is best suited to a given structural setting.

Let $\mathcal{F}$ denote the collection of baseline distributions considered in the study, Γ the set of contamination rates, $\mathcal{C}$ the set of outlier scales, $\mathcal{M}$ the collection of contamination mechanisms, and $\mathcal{N}$ the set of sample sizes. Define the regime space as

$$\mathfrak{R} = \mathcal{F} \times \Gamma \times \mathcal{C} \times \mathcal{M} \times \mathcal{N}.$$

A particular regime is represented by

$$\mathcal{R} = (F_0, \gamma, c, m, n) \in \mathfrak{R},$$

where F_0 is the baseline distribution, γ is the contamination rate, c is the outlier scale, m is the contamination mechanism, and n is the sample size. The distribution induced by $\mathcal{R}$ is denoted by

$$F_{\mathcal{R}} = (1-\gamma)F_0 + \gamma G_{m,c}(F_0),$$

where $G_{m,c}(F_0)$ denotes the contamination distribution generated from F_0 under mechanism m and scale c. Support, skewness, and tail behavior are properties of the resulting distribution $F_{\mathcal{R}}$. For notational simplicity, once a regime is fixed, the induced distribution is written as $F = F_{\mathcal{R}}$.

The learning problem is then to identify a mapping

$$\mathcal{R} \longrightarrow \widehat{\theta}^*_{\mathcal{R}},$$

where $\widehat{\theta}^*_{\mathcal{R}}$ minimizes a prespecified risk criterion within that regime. This formulation is not a literal extension of the finite No Free Lunch theorem to continuous estimation spaces. It follows the same underlying logic: when the candidate data-generating environments contain incompatible structures, optimality is expected to depend on the regime instead of extending universally across all settings [wolpert1997nfl,goldblum2024nfl,wolpert2021metainduction,sterkenburg2021nfl].

C.2 Target functional and admissibility

Conditional optimality is meaningful only after the target has been fixed. Here, the estimand is the population mean functional

$$\theta(F) = \mu_F = E_F[X],$$

whenever $E_F[|X|] < \infty$. This distinction is important under asymmetry, where the mean, median, mode, geometric mean, and harmonic mean describe different features of the distribution. Because squared error loss elicits the mean, MSE comparisons are meaningful here only when all procedures are evaluated as estimators of μ_F [gneiting2011point].

The term admissible is used here in an operational sense and not in the decision-theoretic sense of an estimator being undominated. For validation stage s, let $\mathcal{B}_s$ denote the benchmark registry available at that stage. Define

$$\mathcal{B}_s^{\mathrm{adm}}(\mathcal{R}) = \{T_{b,n} \in \mathcal{B}_s : T_{b,n} \text{ is well defined under } F_{\mathcal{R}}\},$$

with membership further restricted by the benchmark policy specified for stage s. The strongest admissible benchmarks are selected separately for the two risk criteria:

$$T_{b,s,n}^{*,\mathrm{mean}} = \pi_s^{\mathrm{mean}}\left(\mathcal{B}_s^{\mathrm{adm}}(\mathcal{R})\right)$$

and

$$T_{b,s,n}^{*,q} = \pi_s^{q}\left(\mathcal{B}_s^{\mathrm{adm}}(\mathcal{R})\right),$$

where π_s^{mean} and π_s^q denote the prespecified benchmark-selection rules for MSE and $q_{0.95}$, respectively, as described in Section 3.2 of the main text. Thus, strongest refers exclusively to the criterion-specific benchmark selected by the stage-specific comparison policy; it does not denote decision-theoretic admissibility. Every eligible benchmark is evaluated against the same target μ_F.

This choice imposes two constraints. Support admissibility excludes the harmonic and geometric means from regimes in which observations may be non-positive. Target compatibility is more subtle. Even under positive support, the harmonic mean, geometric mean, median, and modal estimators are generally inconsistent for μ_F under asymmetry. They may nevertheless reduce finite-sample risk when used as components of a composite. The search must discourage solutions that appear stable only because they drift toward a different functional. Bias control, dominance control, and benchmark-relative pressure are introduced for this purpose.

C.3 Composite estimators and conditional properties

Let $X_1, \ldots, X_n$ be iid observations from F, with observed sample $x = (x_1, \ldots, x_n)$. Let $T_{1,n}, \ldots, T_{L,n}$ denote a set of base location statistics, and define the L-component probability simplex as

$$\Delta_L = \left\{ w \in [0,1]^L : \sum_{j=1}^{L} w_j = 1 \right\}.$$

For a fixed weight vector $w \in \Delta_L$, the resulting composite estimator is

$$\widehat{\theta}_{w,n}(x) = \sum_{j=1}^{L} w_j T_{j,n}(x).$$

This construction remains interpretable because the estimator is a weighted combination of familiar statistical summaries. Its properties, however, depend on the components receiving positive weight and on the distributional regime in which the composite is used.

Because L is finite and w remains fixed, if every positive-weight component is consistent for the same target μ_F, so that

$$T_{j,n} \xrightarrow{p} \mu_F \qquad \text{as } n \to \infty$$

for every j such that $w_j > 0$, then

$$\widehat{\theta}_{w,n} \xrightarrow{p} \mu_F.$$

This observation states the target-compatibility condition required for consistency; it is not presented as a separate theoretical contribution. The condition is essential. Harmonic, geometric, and modal components are not assumed to be individually consistent for μ_F across general skewed regimes. They enter the composite as auxiliary components that may improve the balance between bias and variance, but their value must be established through comparison with an operationally admissible benchmark.

Breakdown and influence properties also depend on the weights. When the sample mean receives positive weight, the composite will inherit an asymptotic breakdown point of zero. At the functional level, suppose that the functionals underlying the positive-weight components admit influence functions at F and that the regularity conditions required for linearity hold. The influence function of the fixed composite can then be written as

$$\mathrm{IF}(z; \theta_w, F) = \sum_{j=1}^{L} w_j \, \mathrm{IF}(z; T_j, F).$$

For the population mean functional,

$$\mathrm{IF}(z; \mu, F) = z - \mu_F,$$

which is unbounded. Consequently, when $w_{\text{mean}} > 0$ and the remaining positive-weight components have bounded influence functions, the composite influence function remains unbounded. The mean weight scales its contribution but does not make it bounded [hampel1986robust,avellamedina2017influence]. The framework therefore does not assume that combining estimators automatically produces robustness. Instead, the weights reveal the balance among components, while the benchmark gate determines whether that balance is favorable within a particular regime.

The principal risk quantity considered for the composite is mean squared error. Let

$$b_{j,n}(F) = E_F[T_{j,n}] - \mu_F$$

denote the finite-sample bias of component j. Because the weights are fixed, the finite-sample bias of the composite is

$$B_n(w, F) = E_F[\widehat{\theta}_{w,n}] - \mu_F = \sum_{j=1}^{L} w_j b_{j,n}(F).$$

Under iid sampling, the sample mean is unbiased whenever $E_F[X]$ exists and is finite, so

$$b_{\text{mean},n}(F) = 0.$$

The composite bias therefore satisfies

$$|B_n(w, F)| \le (1 - w_{\text{mean}}) \max_{j \neq \text{mean}} |b_{j,n}(F)|.$$

Retaining some weight on the sample mean limits how far the composite can move away from μ_F, although it does not by itself ensure consistency or robustness.

Proposition 1 (Finite-sample MSE dominance region)**.** *For fixed n, suppose that the relevant second moments are finite. Let*

$$V_n(w, F) = \mathrm{Var}_F(\widehat{\theta}_{w,n})$$

denote the finite-sample variance of the composite and let $B_n(w, F)$ denote its finite-sample bias. Then $\widehat{\theta}_{w,n}$ has strictly smaller MSE than $\bar{X}_n$ for estimating μ_F if and only if

$$B_n(w, F)^2 + V_n(w, F) < \mathrm{Var}_F(\bar{X}_n).$$

For the strongest operationally admissible benchmarks selected at stage s, the corresponding conditions are

$$\mathrm{MSE}_{F,n}(\widehat{\theta}_{w,n}) < \mathrm{MSE}_{F,n}\left(T_{b,s,n}^{*,\mathrm{mean}}\right)$$

and

$$q_{0.95,F,n}(\widehat{\theta}_{w,n}) < q_{0.95,F,n}\left(T_{b,s,n}^{*,q}\right).$$

Proposition 1 isolates the sample mean case and clarifies the balance between efficiency and robustness. The benchmark gate evaluates the broader comparison empirically against the criterion-specific comparators selected from $\mathcal{B}_s^{\mathrm{adm}}(\mathcal{R})$ by the prespecified stage-specific policy.
Although elementary, this inequality clarifies when a composite can improve on the sample mean. Under right skewness and heavy tails, $\mathrm{Var}_F(\bar{X}_n)$ may be large. A composite can then accept a small amount of squared bias in exchange for a sufficiently large reduction in variance and still achieve lower MSE.

As F approaches a light-tailed, nearly symmetric regime in which the sample mean is efficient, progressively less variance remains available for a composite to reduce without introducing offsetting bias [lehmann1998point,vandervaart1998asymptotic]. Under such conditions, any successful composite would be expected to assign almost all of its weight to the sample mean. Composite gains should be more plausible in skewed and heavy-tailed families such as Lognormal, Weibull, Inverse Gaussian, and Ex Wald, and less likely in Normal and other nearly efficient regimes. This pattern is a theoretical expectation to be tested by the benchmark gate, not a condition encoded in the selection rule.

The same decomposition explains why performance may not transfer across regimes. Both $B_n(w, F)$ and $V_n(w, F)$ depend on F. Weights learned under a training regime F_{train} may produce different bias and variance under F_{test}. When the distributional structure changes, a favorable reduction in variance can disappear while the bias remains.

This is the mechanism behind the No Free Lunch pattern considered in the study. No single weight vector should be expected to have an MSE dominance region covering several structurally incompatible regimes [sterkenburg2021nfl,goldblum2024nfl]. The selected weights are frozen and evaluated again on locked-unseen regimes and external data. In the language of Proposition 1 this evaluation asks whether the dominance inequality continues to hold under a distribution that played no role in choosing w.

The bias penalty and admissibility mask used during discovery serve related purposes. The penalty discourages weight vectors with large estimated bias, while the mask excludes harmonic and geometric components whenever their support requirements are not satisfied. In those cases, $b_{j,n}(F)$ would not merely be large; the corresponding statistic could be undefined.

A final distinction is important because the composite is later compared with Catoni-type, median-of-means, winsorized, tuned Huber, and trimmed-mean estimators. The citations to

Lugosi and Mendelson concern the broader theoretical literature on robust mean estimation, including median-of-means and trimmed-mean methods; they do not denote a separate estimator in the registry. Under their respective assumptions, Catoni's theoretically calibrated estimator, median-of-means procedures, and certain trimmed-mean constructions admit finite-sample deviation guarantees that keep estimation error small with high probability under heavy tails [catoni2012challenging,lugosi2019mean,lugosi2021trimmed]. These guarantees are not attributed to the implementation-specific Catoni-type variants used here. The composite has no comparable guarantee. Proposition 1 identifies the condition under which a weighted combination can improve, but that condition depends on the unknown quantities $B_n(w, F)$ and $V_n(w, F)$. It characterizes the relevant bias and variance balance without providing an a priori performance bound.

Evidence for a particular composite must come from frozen-weight evaluation under the benchmark gate, locked-unseen regimes, and external error control. Because the behavior of the individual components is known, a deviation bound for a fixed convex combination may be derivable under finite variance. Developing such a bound remains a theoretical extension and is not claimed in the present article.

C.4 Aggregation and evolutionary search over the simplex

The framework is related to estimator aggregation, shrinkage, Bayesian model averaging, and stacking, but it differs in both the statistical object being combined and the final decision rule. James-Stein shrinkage improves estimation of multivariate normal means under squared loss [stein1956inadmissibility,james1961estimation]. Bayesian model averaging combines predictive models through posterior model probabilities [hoeting1999bma], while stacking and the Super Learner combine prediction rules according to cross-validated risk [vanderlaan2007super,claeskens2008model,yang2001adaptive]. Here, the combined objects are location estimators, the target remains fixed at $\mu_F = E_F[X]$, and learning occurs across simulated distributional regimes instead of a single observed prediction dataset. The final rule also permits abstention: a composite is accepted only when it outperforms the strongest admissible benchmark.

Searching for such a composite requires optimization over the simplex Δ_L. The fitness surface is not smooth because it combines quantile-based tail risk, maximum loss, bias, instability, entropy, dominance, and benchmark-relative terms estimated across stochastic Monte Carlo scenarios. Gradient estimation would require additional fitness evaluations and smoothness assumptions that do not fit this objective naturally. Projected gradient and mirror descent methods are not used as the default search procedures, although they remain valuable comparators for future evaluation [beck2003mirror].

The genetic algorithm is suitable for this setting because it relies only on function evaluations, maintains a population of candidate weight vectors, and applies convex crossover and Dirichlet mutation without leaving the simplex [back1996evolutionary,Eiben2015,telikani2021evolutionary,katoch2021review,zha
Its population structure also supports the regime-dependent design. A persistent warm-start bank retains strong weight vectors from earlier regime-specific searches and introduces them into related runs, allowing useful structures to be reconsidered without treating them as predetermined solutions.

This allocation of search effort is conceptually related to successive halving and Hyperband [jamieson2016nonstochastic,li2017hyperband]. The genetic algorithm is not assumed to be uniquely optimal, and a carefully designed CMA-ES procedure with simplex constraints and warm starts would provide a meaningful comparison. Hansen's tutorial offers an accessible account of CMA-ES and provides useful background for understanding how such an alternative search could

be implemented in this setting [hansen2016cma]. The empirical question remains more focused: whether the weights produced by the present search remain competitive once they are frozen and evaluated under held-out and fixed-weight validation.

C.5 Estimator space and evolutionary search

Each individual in the genetic algorithm population is represented by a weight vector $w \in \Delta_L$. The initial population is sampled from a symmetric Dirichlet distribution and supplemented with warm-start candidates retained from previous regime-specific runs. Tournament selection chooses the parent vectors, and convex crossover produces offspring of the form

$$c = \alpha p_1 + (1 - \alpha) p_2, \qquad \alpha \sim U(0, 1).$$

Dirichlet mutation perturbs and renormalizes the weights, ensuring that every candidate remains a valid point on the simplex [Aitchison1982,PawlowskyGlahn2015]. Elitism preserves the strongest candidates, while periodic immigration replaces weak candidates with fresh Dirichlet draws. Reproducibility across parallel runs is maintained through L'Ecuyer CMRG random number streams [lecuyer2002streams].

The initial discovery stage searches over ten learnable components: the mean, median, 20% trimmed mean, harmonic mean, geometric mean, half-sample mode, Parzen kernel mode, Tukey trimean, Huber location, and Tukey biweight location. The expanded discovery stage extends this basis to 26 components by adding winsorized means at 5%, 10%, and 20%; median-of-means estimators with $k \in \{5, 10, 20\}$ blocks; five Catoni-type M-estimators; and Huber tuned locations with

$$k \in \{0.75, 1, 1.345, 1.75, 2\}.$$

The Catoni-type candidates are defined as solutions in θ of

$$\sum_{i=1}^{n} \psi(a(X_i - \theta)) = 0,$$

where

$$\psi(u) = \operatorname{sign}(u) \log\left(1 + |u| + \frac{u^2}{2}\right),$$

and

$$a \in \{0.05, 0.10, 0.20, 0.35, 0.50\}.$$

Here, a is an implementation-specific sensitivity parameter used to define five distinct candidate estimators. The grid was fixed before expanded discovery to span increasingly nonlinear score behavior. It was not designed to reproduce the variance- and confidence-calibrated choice of α derived in Catoni's theoretical analysis. Because $\psi(au) \approx au$ as $a \to 0$, smaller values yield behavior closer to the sample mean, whereas larger values produce stronger nonlinear attenuation of extreme residuals. These candidates are therefore described as Catoni-type estimators and are evaluated empirically under the same frozen benchmark protocol as the remaining components. No finite-sample deviation guarantee from Catoni's theoretically calibrated estimator is attributed to this implementation [catoni2012challenging].

C.6 Data generating families and contamination

Synthetic super populations were generated from six distributional families: Normal, Lognormal, Weibull, Inverse Gaussian, Ex Gaussian, and Ex Wald. These families were selected because they cover symmetry, positive skewness, flexible hazard structures, asymmetric duration tails, exponential reaction time tails, and first passage dynamics [aitchison1957lognormal,weibull1951,chhikara1989inversegaussian,schwarz2001exwald,heathcote2004fitting].

Samples were drawn at sizes

$$n \in \{300, 500, 1000, 2000, 5000\}.$$

Contamination was introduced at rates

$$\gamma \in \{0.00, 0.01, 0.02, 0.05, 0.10, 0.15, 0.20, 0.30\},$$

with outlier scales

$$c \in \{1.5, 3, 4.5, 6, 9, 12, 15, 20, 30\}.$$

Eight contamination mechanisms were considered: upper-tail, lower-tail, symmetric t, point-mass, clustered upper, clustered symmetric, bimodal near, and bimodal far.

Together, these dimensions define the regime structure over which the candidate estimators are discovered and evaluated. Each regime reflects a particular combination of distributional family, sample size, contamination rate, outlier scale, and contamination mechanism.

C.7 Simulation validation, performance criteria, and benchmark gate

Before optimization, the simulation engine was examined through four checks: moment fidelity, contamination injection fidelity, recovery of established statistical benchmarks, and empirical anchoring against real datasets. The benchmark recovery layer verified that the sample mean is efficient under clean Normal sampling, that the median dominates under severe upper-tail contamination, and that Huber location improves on the mean under moderate contamination. These checks follow established recommendations for the design and reporting of simulation studies [law2015simulation,morris2019simulation].

Performance was evaluated using MSE and $q_{0.95}$. MSE summarizes average squared error across the relevant replicates, while $q_{0.95}$ is the 95th percentile of the replicate-level squared-error distribution and captures an upper-tail loss threshold. This second criterion is a quantile risk measure analogous to Value at Risk. Unlike CVaR, it reports the loss quantile itself and does not average losses beyond that threshold [artzner1999coherent,rockafellar2000cvar].

For validation stage s and regime $\mathcal{R}$, the benchmark gate operates on the operationally admissible registry

$$\mathcal{B}_s^{\mathrm{adm}}(\mathcal{R})$$

defined in Section 2 of the main text. The strongest benchmark is selected separately for each risk criterion. Specifically, define

$$T_{b,s,n}^{*,\mathrm{mean}} \in \operatorname*{arg\,min}_{T_{b,n} \in \mathcal{B}_s^{\mathrm{adm}}(\mathcal{R})} \mathrm{MSE}_{F,n}(T_{b,n})$$

and

$$T_{b,s,n}^{*,q} \in \operatorname*{arg\,min}_{T_{b,n} \in \mathcal{B}_s^{\mathrm{adm}}(\mathcal{R})} q_{0.95,F,n}(T_{b,n}),$$

where

$$q_{0.95,F,n}(T_{b,n}) = q_{0.95}\left[\left(T_{b,n} - \mu_F\right)^2\right].$$

The benchmarks minimizing MSE and $q_{0.95}$ need not be the same estimator.

A candidate composite clears the dual benchmark gate only when

$$\mathrm{MSE}_{F,n}(\widehat{\theta}_{w,n}) < \mathrm{MSE}_{F,n}\left(T_{b,s,n}^{*,\mathrm{mean}}\right)$$

and

$$q_{0.95,F,n}(\widehat{\theta}_{w,n}) < q_{0.95,F,n}\left(T_{b,s,n}^{*,q}\right).$$

Thus, a composite that improves $q_{0.95}$ but worsens MSE is rejected, as is a composite that improves MSE while worsening $q_{0.95}$. When both requirements are not satisfied simultaneously, the procedure abstains and retains the relevant criterion-specific benchmark values. The resulting rule is stricter than selection based only on the lowest average loss.

C.8 Initial discovery and fixed-weight confirmation

Initial discovery uses three-fold cross-validation and a multistage allocation of computational resources. The first high-pass filter, HPF1, performs a broad, relatively low-cost scan across configurations and regimes using 60% scenario exposure. HPF2 then reevaluates the more promising signals using 80% exposure. Specialist halving concentrates the remaining computation on the strongest regime-specific candidates, and final discovery decisions use held-out scenarios without retraining.

At the end of discovery, a composite is accepted only when it improves simultaneously on MSE and $q_{0.95}$ relative to the best admissible classical benchmark. Discovery nominates candidate weight vectors, but does not provide the final evidential claim.

Once discovery is complete, the selected weights are frozen and evaluated again without further optimization. Table 1 summarizes the distinction between discovery and fixed-weight confirmation. Confirmation expands the benchmark set to include Catoni-type estimators, median-of-means estimators, winsorized means, tuned Huber estimators, and an equal-weight composite. It also raises precision to $R = 500$ Monte Carlo replicates and $B = 500$ bootstrap resamples, introduces eight additional validation seeds, and evaluates each candidate in both its original regime and a locked-unseen related regime.

The locked-unseen related regime was constructed deterministically from the discovery regime and was never available during HPF1, HPF2, specialist training, specialist validation, or the original held-out gate. Let $\Gamma_{\mathcal{R}}$ and $\mathcal{C}_{\mathcal{R}}$ denote the contamination rates and outlier scales used by the discovery regime, and let $\widetilde{\gamma}_{\mathcal{R}}$ and $\widetilde{c}_{\mathcal{R}}$ denote their medians. The validation values were selected from the prespecified global grids as

$$\gamma^{\mathrm{LU}} \in \underset{\gamma \in \Gamma \setminus \Gamma_{\mathcal{R}}}{\arg\min} \left|\gamma - \widetilde{\gamma}_{\mathcal{R}}\right|, \qquad c^{\mathrm{LU}} \in \underset{c \in \mathcal{C} \setminus \mathcal{C}_{\mathcal{R}}}{\arg\min} \left|c - \widetilde{c}_{\mathcal{R}}\right|.$$

When two grid values were equally close, the smaller value in the sorted grid was selected. The contamination mechanism was kept within the same structural group: upper-tail mechanisms paired `upper_tail` with `clustered_upper`; symmetric mechanisms paired `symmetric_t` with `clustered_symmetric`; bimodal mechanisms paired the near and far bimodal mixtures; and point-mass mechanisms retained `point_mass`. An unused mechanism from the same group was selected when available; otherwise the original mechanism was retained. The distribution family, frozen candidate weights, and sample-size support were unchanged. Thus, relatedness denotes adjacency

under a prespecified contamination-profile rule, not an informal judgment. Exact original and locked-unseen profiles for the confirmed specialists are reported in Appendix A

Relative gains are accompanied by paired bootstrap intervals [efron1993bootstrap]. In the archived configuration, seed-level gate status requires positive point gains for both MSE and $q_{0.95}$; the intervals quantify uncertainty separately. Headline confirmation is limited to candidates that remain stable across the validation seeds and retain positive reported $q_{0.95}$ intervals in the relevant validation mode. This stage separates properties of the frozen estimator from artifacts of the discovery cells.

Table 1: Discovery and fixed-weight confirmation settings. Confirmation uses the frozen discovery weights and performs no additional genetic algorithm search.

Component	Discovery	Fixed-weight confirmation
Distribution families	6	Focal validation regimes from discovery
Scenario universe	576 per family	Original and deterministically constructed locked-unseen related regimes
Genetic algorithm basis	10 components	Frozen discovery weights; no new genetic algorithm search
Discovery seeds	101, 202	Not rerun
Validation seeds	–	303, 404, 505, 606, 707, 808, 909, 1010
Monte Carlo replicates	$R = 40$	$R = 500$
Bootstrap replicates	$B = 100$	$B = 500$
Benchmarks	Classical admissible set	Catoni-type, MOM, winsorized, tuned Huber, equal weight composite
Final rule	Benchmark gate	Expanded benchmark gate with paired bootstrap intervals

C.9 Expanded-basis rediscovery and post-discovery validation

The expanded-basis stage reopens discovery with modern robust estimators available from HPF1 onward instead of adding them only after candidate selection. The genetic algorithm now learns weights over the complete 26-component registry defined in Section 3.1 of the main text, including Catoni-type, median-of-means, winsorized, and tuned Huber estimators.

The regime-first architecture remains unchanged. HPF1, HPF2, specialist halving, and the held-out benchmark gate retain the same roles, allowing the change in the estimator basis to be examined without replacing the broader discovery structure. HPF1 continues to use 60% scenario exposure, while HPF2 increases from 80% in the initial stage to 90% in the expanded stage.

Table 2 summarizes the design of the two discovery stages.

Inheritance between the two stages is controlled. Only weights that pass the initial fixed-weight confirmation are allowed to enter the expanded-basis stage as protected warm-start candidates. In the completed run, this condition applies to CV-019 and CV-010. Their inclusion preserves information from the previous stage, but does not grant automatic acceptance. They retain external prior provenance and must compete again under the 26-component gate.

After expanded discovery, candidates are evaluated with frozen weights through the post-discovery fixed-weight validation, with $N_{\mathrm{EST}} = 26$. The resulting evidence taxonomy assigns each candidate to one of four reporting categories: transfer specialist, discovery-supported local candidate, near-gate candidate, or negative-control success. These categories summarize the strength and scope of the evidence but do not alter the decisions produced by the benchmark gate.

Table 2: Design of the two discovery stages. The expanded-basis stage includes modern robust estimators directly in the discovery space instead of introducing them only as subsequent competitors.

Design element	Initial discovery	Expanded-basis rediscovery
Discovery gate	Classical admissible set	Expanded modern set active from HPF1
Genetic algorithm composite basis	10 estimators	26 estimators, including Catoni-type, MOM, winsorized, and Huber tuned components
HPF1 scenario exposure	60%	60%
HPF2 scenario exposure	80%	90%
Warm starts	Dirichlet plus regime bank	Dirichlet, regime bank, and confirmed Lognormal weights
Discovery seeds	101, 202	101, 202
Audits	Post-discovery diagnostics	Profile, global, and interfamily audits integrated into the close of discovery
Post-discovery validation	Expanded fixed-weight confirmation	post-discovery fixed-weight validation ($N_{\mathrm{EST}} = 26$)

The allocation of computational resources follows the logic of successive halving and Hyperband [jamieson2016nonstochastic,li2017hyperband]. The initial stage establishes the discovery framework and evaluates composites learned under classical benchmark pressure. The expanded stage then tests whether introducing modern robust components from the beginning changes which regions of the simplex become accessible to the search. Together, the two stages provide a connected examination of conditional optimality, not two separate studies.

C.10 Confirmation and abstention audit design

Fixed-weight confirmation was challenged through four complementary audits: a robustness audit based on the frozen component weights and gross-error sensitivity; a bootstrap audit focused on nonsmooth median and modal components; a Benjamini–Hochberg correction across confirmation candidates; and a split-sample check separating candidate selection from performance reporting across the eight validation seeds. These audits did not modify the benchmark gate or the evidence taxonomy.

A complementary abstention audit examined whether benchmark retention could reflect incomplete exploration of the simplex. For each of 17 benchmark-retained regimes and both validation modes, 4000 weights were sampled from a symmetric Dirichlet distribution with concentration parameter $\alpha = 0.30$ and evaluated using the same simulator, admissibility rules, benchmark registry, definition of $q_{0.95}$, $R = 500$ Monte Carlo budget, and eight validation seeds. A regime and mode cell confirmed abstention when no sampled vector cleared both gates at all eight seeds. The frozen Weibull specialists served as positive controls, while randomly passing vectors were retained only as future search targets and were not promoted to confirmed discoveries.

The complete audit procedures, including the influence-function interpretation, caveat for nonsmooth bootstrap components, multiplicity rule, split-sample partition, and random-search protocol, are reported in Repository Appendix B.

C.11 External validation design

External validation examined whether the frozen candidates transferred beyond the synthetic super populations. All weights were applied exactly as discovered and validated, without retraining, adaptation, or further optimization on the public datasets.

Because the population mean is not known independently in observational public data, the external target was operationalized as the full-sample mean of each dataset. Each specialist was evaluated according to how accurately it recovered that reference mean from repeated subsamples. The external analysis therefore assesses sampling variability and robustness relative to the full-sample reference. It does not constitute validation against a population mean known independently of the observed dataset.

Tail observations were identified endogenously within each public data marginal, and no synthetic contamination was added. The implementation used the unscaled median absolute deviation

$$\mathrm{MAD}_0(x) = \mathrm{median}_i \, |x_i - \mathrm{median}_j(x_j)|$$

and calculated

$$z_i = \frac{x_i - \mathrm{median}(x)}{s(x)},$$

where $s(x) = \mathrm{MAD}_0(x)$. If this scale was non-finite or non-positive, the implementation used $\mathrm{IQR}(x)/1.349$, followed by the standard deviation if necessary. Depending on the prespecified contamination direction, the empirical tail pool contained observations with $z_i \geq 3$, $z_i \leq -3$, or $|z_i| \geq 3$. The threshold was fixed before the external battery was completed. It is a prespecified operational cutoff, not a claim that $z = 3$ is universally optimal; lower MAD-based cutoffs have also been recommended in the literature [leys2013outliers]. No parallel analysis at 2.5 or 4 was conducted, so the external counts and profile-specific conclusions are conditional on the $z = 3$ rule. Because the unscaled form of MAD was used, numerical comparisons with published cutoffs depend on the scaling convention. In skewed marginals, a flag identifies an observation as operationally irregular under this rule; it does not establish a distinct contamination process [leys2013outliers,berger2021rtoutliers,miller2023rtoutliers]. The analysis therefore tests transfer conditional on an empirical profile, not performance under arbitrary contamination.

Public sources were assembled while balancing profile quotas and source caps. After loading and eligibility screening, dataset-level evaluations were constructed from the retained sources. These flow counts are reported in Section 4 of the main text. The primary external claim uses the independent parent dataset as its inferential unit, not each generated comparison. Each dataset-level evaluation could generate multiple eligible comparisons across frozen candidates, empirical profiles, validation modes, and subregimes. After profile matching and eligibility screening, the external battery produced 809 dataset-level comparison rows for multiplicity control.

Three prespecified safeguards governed the external analysis. First, a profile gate restricted primary evaluation to specialist and dataset pairs whose empirical distributional profile matched the family associated with the specialist. Nonmatching pairs were retained only as cross-regime audit rows and were excluded from the primary evidence count. When a dataset did not match a validated specialist family, the framework retained the corresponding benchmark.

Second, the primary comparator was selected through a benchmark policy prespecified by empirical profile. A best-across-all-estimators oracle was calculated separately as a secondary audit, but it did not replace the profile-specific primary benchmark and was not used to define the principal external claims.

Third, uncertainty and multiplicity control were applied to the exact reported gain. Paired bootstrap intervals were computed from the paired specialist and benchmark losses, and the original

Benjamini–Hochberg procedure [benjamini1995controlling] was applied at the dataset level across primary profile-eligible, in-regime comparisons. For each eligible candidate, dataset, and mode comparison, the point gate required positive mean and $q_{0.95}$ gains, and the paired bootstrap interval for the prespecified reported gain had to remain above zero. A strict external win also had to survive the dataset-level false discovery rate correction. The broader closure-principle proposal of Xu et al. [xu2025closure] was not used to define, tune, or justify the correction implemented in this study.

A saturation cap limited each dataset to no more than ten countable wins. Additional within-dataset evaluations were retained as depth probes, explicitly tagged, and excluded from the count of independent datasets. This rule reduces the risk of pseudoreplication by preventing repeated evaluations of the same parent dataset from inflating the reported extent of independent external evidence [hurlbert1984pseudoreplication,aarts2014solution].

External evidence was summarized along two complementary dimensions. Independent dataset coverage was defined as the number of parent datasets on which at least one profile-eligible candidate achieved a strict win supported by its confidence interval and survived false discovery rate control. Within-dataset stability was defined as the number of corrected subregime confirmations observed within those parent datasets.